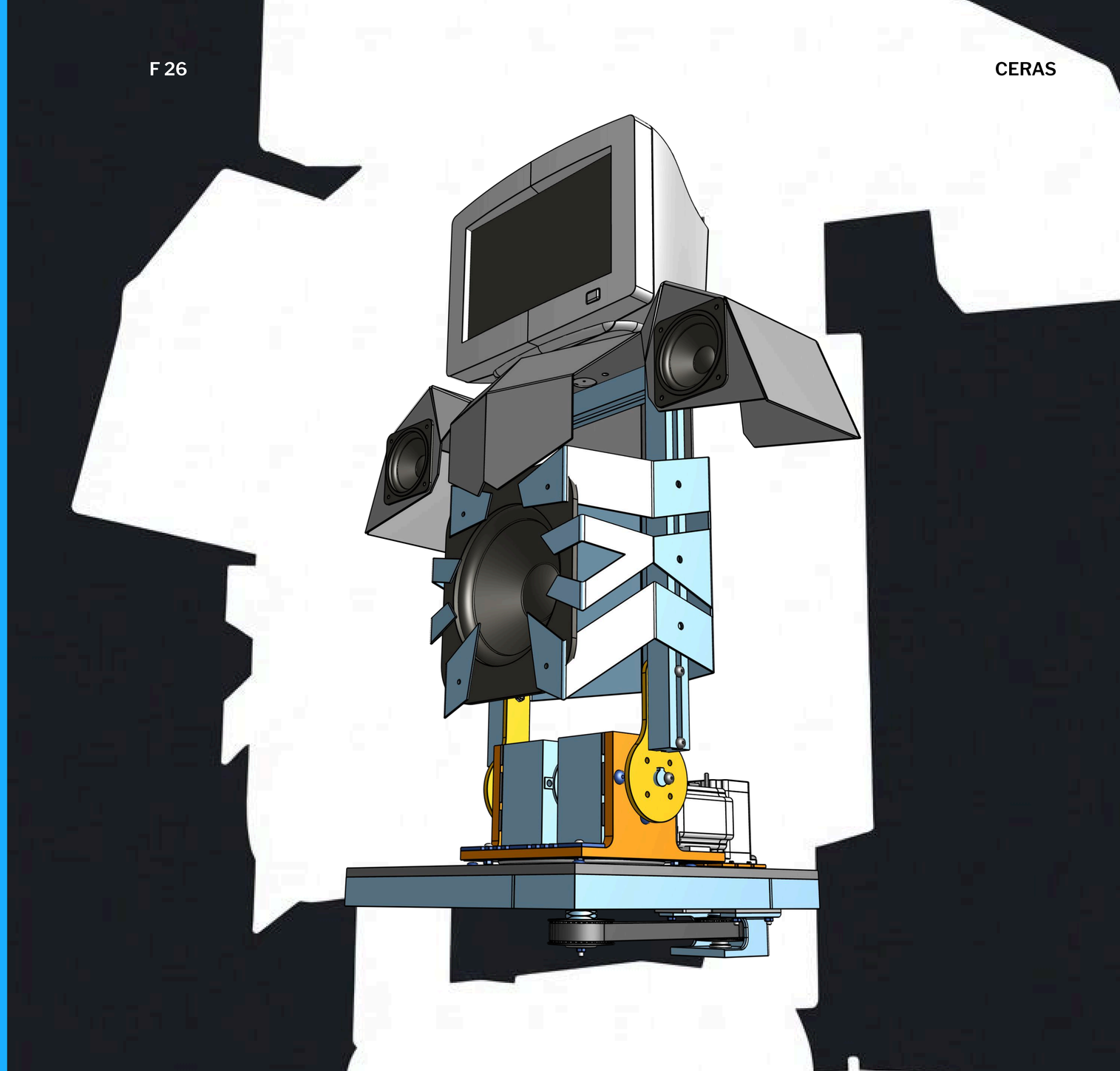
CERAS

CREATIVE ENGINEERING & ROBOTICS ARTS STUDIO



WHAT IS CERAS?

CERAS is a student-led robotics team exploring the intersection of engineering and collaborative art.

Our goal: create robots that don’t just automate tasks, but interact and create with people.

OUR TEAM

- 40 active members
- ME, EE, CE, CS, Physics, Music, Comms
- Heavy involvement at Northwestern's numerous robotics labratories
- Industry experience in robotics, ML, embedded, research & mechatronics
- All VERY passionate about building robots

formlabs

Milwaukee

HAND
HUMAN AUGMENTATION VIA DEXTERITY

Medtronic
Hugo™ RAS System

Goldman Sachs

IBM

EMBODIED SYSTEMS LAB

mundane humanoid robotics

ROBOTICS

APL
JOHNS HOPKINS
APPLIED PHYSICS LABORATORY

Argonne NATIONAL LABORATORY

P&G

清華大學
Tsinghua University

启元机器人PRiME3OT

LEADERSHIP



BRADY LIN
CO-LEAD & HEAD OF CONTROLS

Electrical Engineering ‘28
Mundane — Designed humanoid robot teleoperation controls stacks
Formlabs — Mechatronics integration for cartesian robots



AMANDA DIAO
CO-LEAD & HEAD OF MECHANICAL

Mechanical Engineering ‘28
Medtronic — Built maneuvering mechanisms for surgical robots
Marks — Consumer product mechanical design & prototyping



JONAH KIM
HEAD OF AUTONOMY

Integrated Sciences ‘28
JHU APL — Applied deep learning for missile defense
Palantir — Incoming ML & Automations Intern



PETER WANG
HEAD OF DESIGN

Manufacturing & Design ‘27
Tsinghua Future Lab— Designed dexterous humanoid hands
Primebot — Engineered linkages for Chinese humanoid robots



SIWON RYU
HEAD OF INTEGRATION

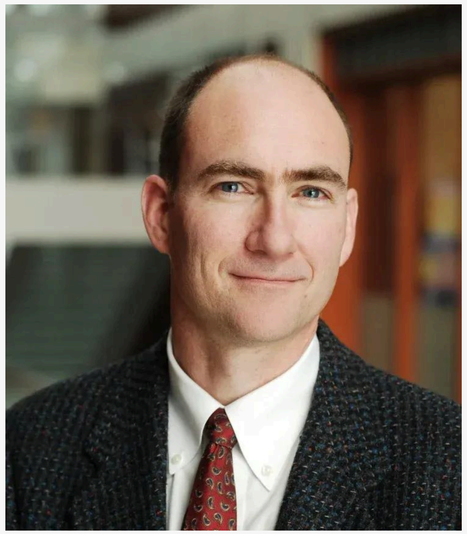
Computer Science ‘27
Apple — Agentic automation for hardware reliability testing
Milwaukee — Streamlined factory automation processes

ADVISORY BOARD



BRENNA ARGALL
TECHNICAL ADVISOR

AIMBE Fellow — Recognized for advances in assistive robotics
Shirley Ryan Ability Lab — Pioneers human-robot collaboration



EDWARD COLGATE
PRINCIPAL ADVISOR

Founded Cobotics — Invented collaborative robots (Cobots)
NSF HAND Center — Leads research in robotic dexterity



MATTHEW ELWIN
ACADEMIC ADVISOR

Directs MS in Robotics — Leads NU's robotics master's program
Wrote the PIC32 Book — a go-to text on embedded mechatronics



NICHOLAS MARCHUK
BUSINESS ADVISOR

Founded nLab: Ed-tech that raised 500k from VCs and Kickstarter
Mechatronics Lab Head — Leads hands-on robotics at Northwestern

"THE UNDERBELLY OF THE MUSIC IS COLLABORATION AND TRUST."

Yo-Yo Ma

World-renowned cellist & cultural advocate

AT CERAS, WE BELIEVE IN

ROBOTS THAT CREATE WITH US.

Music is one of the purest forms of collaboration: listening, responding, and creating together.

Music is our first stage.

A foundation for human–robot collaboration beyond performance.

WHY WE START WITH MUSIC

COLLABORATION YOU CAN HEAR.

Listen.

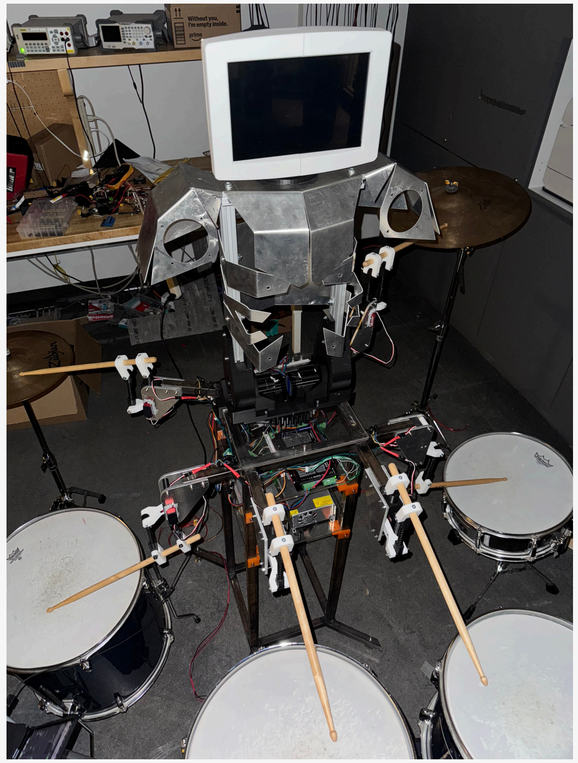
Understand another person's timing and intent.

Respond.

Adjust as the interaction changes.

Create together.

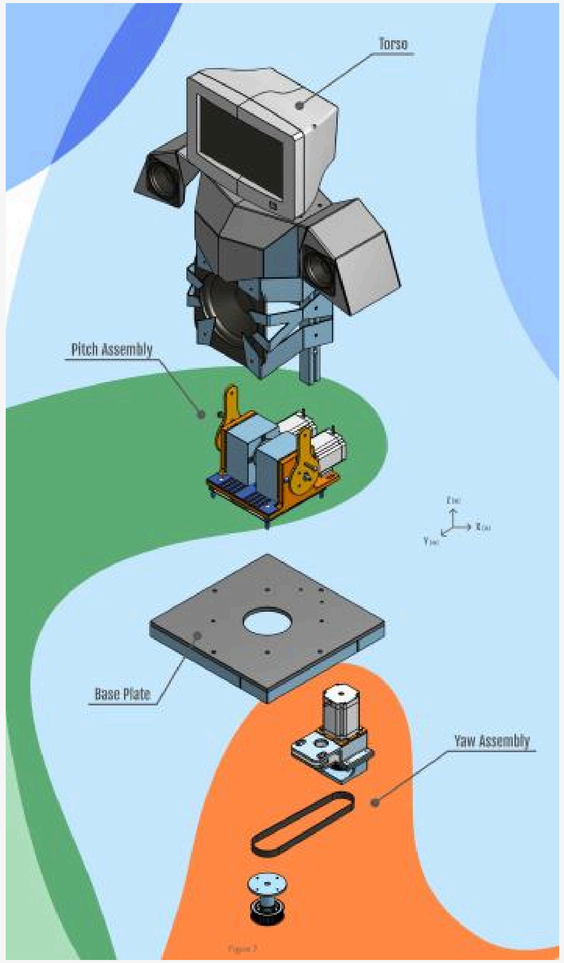
Make something neither participant creates alone.



WHAT WE'VE DONE

- Built a 6 arm robotic system capable of playing an acoustic drum set in under 2 academic quarters
- 100+ custom moving parts made using CNC, 3D printing, and welded sheet metal
- Custom PCBs & electromechanical stacks
- Successfully performed in many concerts on campus







Demo of drumming sequence.

Demo of torso in action

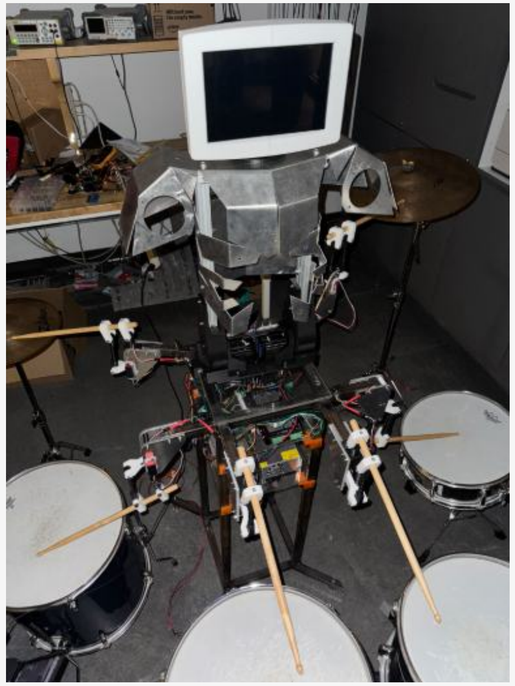


WE'RE GOING HUMANOID

From playing rhythms to playing together.

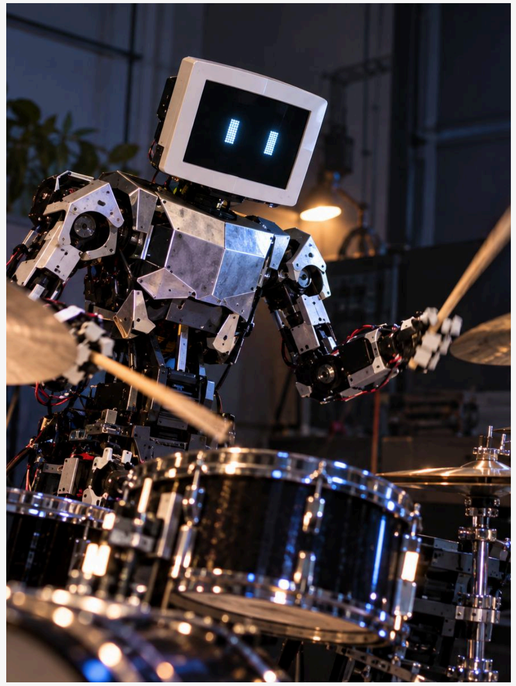
What we've proven:
A robot that performs.

- Built a dedicated drumming system
- Catches attention & performed in concerts
- A foundation to build on



Our next chapter:
Collaborative humanoid musician.

- Learn through hands-on teaching
- Jam and improvise with people
- Develop precise 7-DOF arm motions
- Extend collaboration beyond music



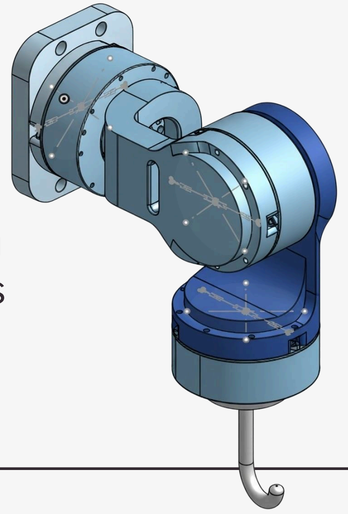
WHAT WE'VE DONE THE PAST 3 WEEKS

Members onboarded. Proof projects completed. Ready for the custom-arm CAD sprint.

Mechanical Track:
All members individually completed a 3-DOF robotic shoulder design around selected actuators and validated that the system could move as intended.

Mechanical work completed:

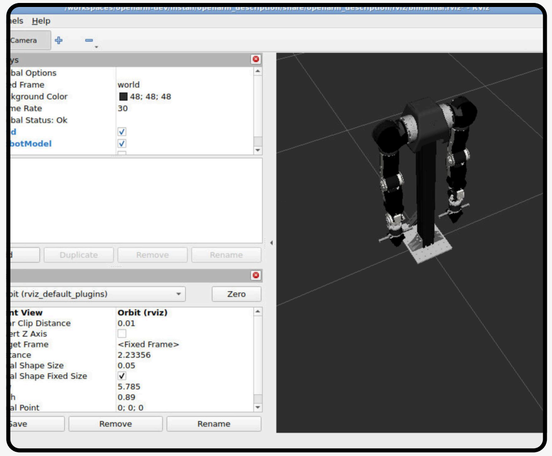
CAD assemblies and arm design
URDF / FK / IK implementation
Load analysis and actuator selection
Stress, deflection, and safety checks



Software Track:
Built and tested a robotics workflow using OpenArm and Kscale-style models. We verified RobStride actuator drivers and simulated robot movements.

Software workflows validated:

ROS 2 integration in MuJoCo
URDF visualization in RViz
Forward / inverse kinematics
Simulation and robot control



THIS YEAR: OUR OWN COLLABORATIVE ARMS

GOAL: Develop our own arm mechanisms and robotics software in parallel, with precise haptics and imitation learning designed for human collaboration.

FALL

DESIGN & VALIDATE

- Run the custom-arm CAD sprint
- Validate mechanisms and manufacturing plans.
- Simulate drumming sequences with ROS2 in MuJoCo
- Validate software on OpenArm

WINTER

BUILD & GROW

- Manufacture and assemble arm prototypes.
- Test software on our custom hardware
- Heavily expand recruitment around active projects.

SPRING

INTEGRATE & DEMONSTRATE

- Complete the pair of arms.
- Demonstrate drumming alongside people
- Develop HRI stack, and do selected manipulation tasks.
- Learn to drum with RL

FROM ARMS TO A FULL HUMANOID

YEAR 2

TORSO, NECK & INTEGRATION

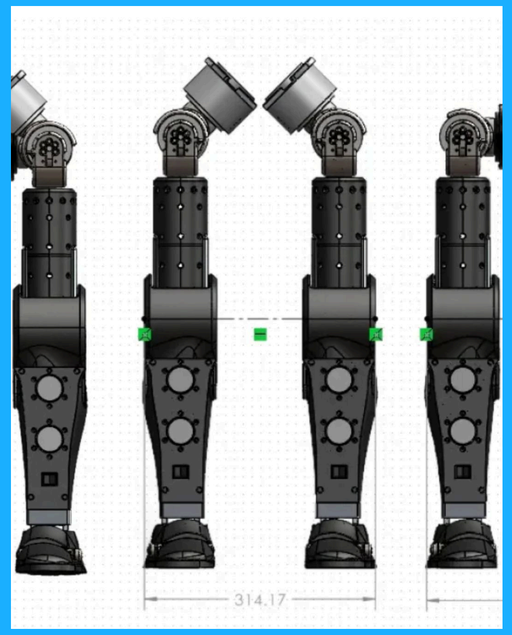
- Build the torso and neck
- Integrate sensing and learned behaviors across the upper body
- Begin research on legs and locomotion



YEAR 3

LEGS & BROADER COLLABORATION

- Build and integrate the legs
- Develop whole-body coordination
- Extend beyond music into tasks where robots learn from and work alongside people



THANK YOU!